

Statis Statistical Suite

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Web apps

Web app	Focus	URL
Statis	Frequentist analysis of datasets: tests, regression, survival, meta-analysis, machine learning	rasan17.github.io/Statis
Statis-Gravity	Study design, descriptive and comparative statistics, diagnostics, multivariate EDA, teaching simulations	rasan17.github.io/statis-gravity
Bayes-Estimation	Bayesian estimation for proportions, means and diagnostic tests	rasan17.github.io/Bayes-Estimation

Statistical tests by web app

Tests are ordered from study design through description, comparison, modelling and synthesis, to Bayesian methods and teaching tools. A tick shows which app offers each one.

Category	Test or procedure	Statis	Statis-Gravity	Bayes-Estimation
1. Study design	Sample size and power: two independent means (t-test)		✓	
1. Study design	Sample size and power: paired means (paired t-test)		✓	
1. Study design	Sample size and power: two proportions (χ^2 / Fisher's exact)		✓	

Category	Test or procedure	Statis	Statis-Gravity	Bayes-Estimation
1. Study design	Post-hoc (achieved) power		✓	
1. Study design	Simple randomisation		✓	
1. Study design	Block randomisation (balanced allocation)		✓	
2. Descriptive	Descriptive statistics (n, mean, median, mode, SD, variance, range)	✓	✓	
2. Descriptive	95% CI for the mean		✓	
2. Descriptive	Skewness and kurtosis		✓	
2. Descriptive	Outlier detection (Tukey's fences)		✓	
2. Descriptive	Frequency histogram, box-and-whisker and violin plots		✓	
2. Descriptive	Pivot tables	✓		
3. Assumption checks	Shapiro-Wilk normality test	✓		
3. Assumption checks	Jarque-Bera normality test		✓	
3. Assumption checks	Levene's test (equality of variances)	✓	✓	
3. Assumption checks	Box-Cox transformation	✓		
4. Two groups	Student's t-test (independent, equal variances)	✓	✓	
4. Two groups	Welch's t-test (independent, unequal variances)	✓	✓	
4. Two groups	Paired t-test	✓	✓	
4. Two groups	Mann-Whitney U test	✓	✓	

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4. Two groups	Wilcoxon signed-rank test	✓	✓	
4. Two groups	Cohen's d	✓	✓	✓
5. Three or more groups	One-way ANOVA	✓	✓	
5. Three or more groups	Welch's ANOVA	✓	✓	
5. Three or more groups	Kruskal-Wallis H test	✓	✓	
5. Three or more groups	One-way repeated-measures ANOVA		✓	
5. Three or more groups	Friedman test		✓	
5. Three or more groups	Post-hoc contrasts (Tukey HSD, Games-Howell, Dunn, Bonferroni)		✓	
5. Three or more groups	Effect sizes (η^2 , ε^2)	✓		
5. Three or more groups	MANOVA (Wilks' Λ) with univariate follow-up	✓		
5. Three or more groups	ANCOVA (homogeneity of slopes, partial η^2)	✓		
6. Categorical data	Pearson's χ^2 test	✓	✓	
6. Categorical data	Yates' continuity correction	✓	✓	
6. Categorical data	Fisher's exact test	✓	✓	
6. Categorical data	Cramér's V	✓		
6. Categorical data	Odds ratio	✓	✓	✓
6. Categorical data	Relative risk	✓	✓	✓
6. Categorical data	Risk difference (absolute risk reduction)	✓		✓
6. Categorical data	Number needed to treat		✓	✓

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6. Categorical data	Wilson score 95% CI for a proportion		✓	
6. Categorical data	Cohen's h			✓
7. Correlation and linear regression	Pearson's r	✓	✓	
7. Correlation and linear regression	Spearman's ρ	✓	✓	
7. Correlation and linear regression	Kendall's τ -b	✓		
7. Correlation and linear regression	Linear (OLS) regression with R^2	✓	✓	
7. Correlation and linear regression	Regression diagnostics (Breusch-Pagan, residual normality, curvature check)	✓		
8. Generalised linear models	Logistic regression (logit or probit link)	✓		
8. Generalised linear models	Firth's penalised-likelihood correction	✓		
8. Generalised linear models	Poisson regression with overdispersion check	✓		
8. Generalised linear models	Relative-risk regression (log-binomial, modified Poisson fallback, bootstrap SE)	✓		
8. Generalised linear models	Likelihood-ratio test, McFadden's pseudo- R^2 , confusion matrix	✓		
9. Survival analysis	Kaplan-Meier estimate with Greenwood variance and median survival	✓		
9. Survival analysis	Log-rank test (two groups)	✓		

Category	Test or procedure	Statis	Statis-Gravity	Bayes-Estimation
9. Survival analysis	Life table	✓		
10. Diagnostic accuracy	Sensitivity, specificity, PPV, NPV		✓	
10. Diagnostic accuracy	Likelihood ratios (LR+, LR-)		✓	✓
10. Diagnostic accuracy	Diagnostic odds ratio			✓
10. Diagnostic accuracy	Post-test probability and Fagan nomogram			✓
10. Diagnostic accuracy	ROC curve and AUC	✓ (ML classifiers)	✓	
10. Diagnostic accuracy	Optimal cut-off (Youden's J)		✓	
11. Meta-analysis	Mantel-Haenszel fixed-effect pooling (OR, RR, RD)	✓		
11. Meta-analysis	DerSimonian-Laird random effects	✓		
11. Meta-analysis	Hartung-Knapp-Sidik-Jonkman adjustment	✓		
11. Meta-analysis	Heterogeneity (Cochran's Q, I ²)	✓		
11. Meta-analysis	Forest plot; funnel plot with Egger's test	✓		
12. Causal and multivariate	Propensity score matching (caliper, SMD balance, ATT)		✓	
12. Causal and multivariate	Principal component analysis (PCA)		✓	
12. Causal and multivariate	Multiple correspondence analysis (MCA)		✓	

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12. Causal and multivariate	Factor analysis of mixed data (FAMD)		✓	
13. Machine learning	Supervised: logistic/softmax, decision tree, random forest, k-NN, neural network	✓		
13. Machine learning	Clustering: k-means, hierarchical (Ward), Gaussian mixture (EM), DBSCAN, silhouette	✓		
14. Bayesian	Two proportions (beta-binomial): posterior RR, OR, ARR; P(treatment > control); ROPE			✓
14. Bayesian	Bayes factor for two proportions (Savage-Dickey)			✓
14. Bayesian	Two means (BEST): $\Delta\mu$, 95% HDI, CLES, variance ratio, ROPE			✓
14. Bayesian	JZS Bayes factor (Cauchy prior) for two means			✓
14. Bayesian	Single proportion (beta-binomial) vs benchmark			✓
14. Bayesian	Single mean (normal-normal) vs benchmark			✓
14. Bayesian	Bayes' theorem and odds updating (unit-square visualisation)		✓	
15. Teaching simulations	Probability distributions (normal, t, log-normal, exponential, bimodal, uniform, Poisson, χ^2)		✓	
15. Teaching simulations	Central limit theorem simulation		✓	

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15. Teaching simulations	Convergence of t to the standard normal		✓	
15. Teaching simulations	Two-sample overlap and α boundary		✓	
15. Teaching simulations	Power ($1 - \beta$) and type II error visualiser		✓	

Notes

- The listing reflects the public pages of each app as read on 3 October 2026.
- Statis states that post-hoc pairwise comparisons, Cox proportional hazards and negative-binomial regression are not yet implemented.
- In Statis, ROC and AUC appear only within the machine-learning module, for two-class classifiers.

Sources

- [Statis](#)
- [Statis-Gravity](#)
- [Bayes-Estimation](#)